	1	2	3	4	5
A	100			20	
B	150			10	
P	–	500			500
Q	–				

4

fractional yield = $\frac{\text{moles desired product}}{\text{moles LR fed}}$

Selectivity = $\frac{\text{moles B converted into P}}{\text{moles B fed}}$

mixer

$P_{in} = P_{out} \Rightarrow P_{out} = 500$ mol/min

mixer DOF $(10+0) - (1+1+0+4) = 4$

$A_2 = 100 + A_5$

$B_2 = 150 + B_5$

$Q_2 = Q_5$

reactor

$A_3 = A_2 - \xi_1$

$B_3 = B_2 - \xi_1 - \xi_2$

$P_3 = 500 + \xi_1$

$Q_3 = Q_2 + \xi_1$

good

total moles are not conserved

fractional yield = $\frac{P_4}{100 \text{ mol A}}$ =

fractional selectivity = $\frac{\xi_1}{150 \text{ mol B}}$ =

Splitter

$A_3 = A_4 + A_5$

$A_3 = 20 + A_5$

$P_3 = P_4 + 500$

$B_3 = 10 + B_5$

$Q_3 = Q_4 + Q_5$

a good start
see solution

$A_3 = .split\ A_3 + (1 - .split) A_3$

$A_3 = 20 + (1 - .split) A_3$

$P_3 = .split\ P_3 + 500$

$B_3 = 10 + (1 - .split) B_3$